

Shikonin mediated anti-acute lung injury and hemostatic effects via inhibiting uPA

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ARTICLE INFO

Keywords:

Shikonin
Acute lung injury
Hemostasis
uPA
Macrophages

ABSTRACT

Shikonin is the main active ingredient in *Arnebiae Radix*, an herbal medicine with anti-inflammatory and hemostatic effects, yet which mechanism remains unveiled. Urokinase-type plasminogen activator (uPA) not only regulates fibrinolysis, but also modulates macrophage invasion, matrix remodeling and adhesion. This study examined whether uPA participated in the anti-inflammatory activity of Shikonin on LPS-induced mice acute lung-injury (ALI) and RAW264.7 models, and investigated whether Shikonin mediated a hemostatic effect by inhibiting uPA in animal serum. The results showed that Shikonin ameliorated inflammation and reduced uPA in the lung of ALI mice. Proteomic results revealed that Shikonin markedly decreased uPA, MMP-9 and pro-inflammatory cytokines in macrophages. Besides, Shikonin decreased serum uPA and promoted blood coagulation. Collectively, current data suggested that Shikonin inhibited uPA, which possibly associated with its anti-inflammatory and hemostatic effects.

1. Introduction

Shikonin is the major representative naphthoquinone component of the herbal medicine *Arnebiae Radix*, the dried root of *Arnebia euchroma* (Royle) Johnston. or *Arnebia guttata* Bunge. *Arnebiae Radix* is reported to have multiple bioactivities, including circulation invigorating, anti-inflammation, antineoplastic, bacteriostatic and hemostatic effects (Zhang et al., 2018). Shikonin significantly inhibited LPS-induced macrophage aggregation and neutrophil infiltration, reduced pro-inflammatory factors (TNF- α , IL-1 β , and IL-6), and inhibited the NF- κ B pathway to exert anti-inflammatory effects (Liang et al., 2013; Mondino & Blasi, 2004; Yang et al., 2018). Shikonin can also reduce the level of inflammatory factors and promote burn wound repair by regulating M1/M2 polarization (Chu et al., 2019; Shu et al., 2022; Tao et al., 2022). However, it remains elusive whether there is a communal mechanism enable the various of activities.

It is known that the fibrinolytic system not only regulates blood coagulation but also associates with inflammation, wound healing and tumor invasion. In this process, urokinase-type plasminogen activator

(uPA), the key serine protease with high specificity, activates fibrinogen alone or in combination with its receptor (uPAR). When bound to uPA, uPAR promotes the enzymatic activation of fibrinolytic enzymes and plays a role in a range of cell migration and invasive metastasis processes by participating in the degradation of the basement membrane or extracellular matrix (ECM) (Bharadwaj, Holloway, Miller, & Waisman, 2021; Xiao et al., 2023). Furthermore, in macrophage, uPA recognized by uPAR activates multiple intracellular inflammatory pathways and uPA activity is a central regulator of 3D invasion, matrix remodeling and adhesion (Fleetwood et al., 2014; Kim, Lee, Choi, Yoo, & Yang, 2012). A recent study showed that uPAR interferes with innate immunity response via TLR4 and this mechanism represents a potentially important target in inflammation and sepsis therapy (Chu et al., 2019; Kiyani et al., 2020).

Acute lung injury (ALI) is an acute, progressive, hypoxic respiratory insufficiency or respiratory failure mostly associated with acute and severe inflammation in lungs (Kor et al., 2016; Patel, Wilson, O'Dea, & Takata, 2013). The disease is likely to develop into acute respiratory distress syndrome (ARDS) if not controlled in time, which may lead to

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multi-organ failure. Therefore, with its high morbidity, ALI is regarded as a common critical clinical condition that poses a serious threat to human health (Butt, Kurdowska, & Allen, 2016). However, effective treatments of ALI are still insufficient. The level of uPA is elevated in pathological settings such as ALI, which induced phosphorylation of eNOS, reduces pulmonary arterial contractility, and increases permeability (Nassar et al., 2011).

Shikonin has protective effects against both lipopolysaccharide and hypoxia-induced lung injury in mice (Liang et al., 2013). It can attenuate inflammation via inhibiting macrophages M1 polarization (Yuan et al., 2014). Besides, Shikonin exerts strong antineoplastic effect on various types of cancer and has a potent wound-healing property. Both of these bioactivities cohere with the clinical application of *Arnebiae Radix*, an herb which in Chinese medical theory is known for promoting blood circulation, removing blood stasis, and stopping bleeding. It has been documented that uPA, as a plasminogen activator, plays an important role in hemostasis and inflammation (Gonias, 2021). Therefore, we wondered whether an inhibitory effect on uPA is related to both anti-inflammatory and hemostatic effects of Shikonin. This experiment was conducted to provide data supporting the clinical application of *Arnebiae Radix*.

2. Materials and methods

2.1. Animals and cells

C57BL/6J mice (male, weighted 25–30 g) were purchased from Guangdong Yaokang Biotechnology Co., LTD. (License No. SCXK, Guangdong 2020-0054). Kunming mice (male and female, weighted 25–35 g) were purchased from the Animal Experiment Center, Guangzhou University of Chinese Medicine (license No. SCXK, Guangdong 2019-0047). Healthy rabbits (male and female, weighted 3.0–3.5 kg) were obtained from Xinhua Animal Farm, Huadu District, Guangzhou (License No. SCXK, Guangdong 2019-0023). RAW264.7 cells were provided by the Cell Resource Center, Shanghai Institutes for Biological Sciences, Chinese Academy of Sciences. RAW264.7 cells were cultured in DMEM medium containing 10 % fetal bovine serum and 1 % penicillin–streptomycin.

2.2. Animal treatment

C57BL/6J mice were randomly divided into five groups with eight animals in each group, namely, normal group, ALI model group, and Shikonin (purchased from Nanjing Daosf Biotechnology Co., LTD, HPLC $\geq$ 98 %, 517-89-5) groups. The animals in Shikonin groups were pre-administered with Shikonin 4 mg/kg, 8 mg/kg or 16 mg/kg (dissolved in the solvent of 10 % DMSO, 15 % olive oil, 5 % ethanol and 70 % PBS) intraperitoneal injection at the volume of 0.01 mL/g body weight. Mice in the normal and model groups were given the same volume of solvent. Two hours later, LPS (5 mg/kg, dissolved in PBS) was i.p. injected to establish a mouse model of ALI, while animals in control group received same volume of PBS. Twelve hours after modeling, Shikonin was administered again following previous protocol. Twelve hours later, animals were euthanized to collect samples for analysis. The lung tissues collected from mice were dissected into four parts for staining, real-time fluorescence quantitative PCR, ELISA, and Western blot analysis respectively. To evaluate the protection of Shikonin on ALI mice, the lung tissues were dehydrated, embedded in paraffin, sliced into 4 μ m sections, stained with H&E, and then observed under a microscope (Shanghai Lai's Electronic Technology Co., LTD, BX53).

2.3. Effects of Shikonin on white blood Cell (WBC), monocyte (Mon) and neutrophil granulocyte (Gran) proportion in peripheral blood of ALI mice

To measure the count of WBC, Mon and Gran in the peripheral blood of mice, 20 μ L of whole blood was add to 20 μ L 0.1 % heparin sodium

(Solarbio, H8060) and detected on a veterinary fully automated whole blood analyzer (Shenzhen Mindray Bio-Medical Electronics Co., Ltd. BC-2800Vet).

2.4. Effect of Shikonin on inflammatory factors and uPA content in lung tissue of ALI mice

Mice lung tissues (0.03–0.04 g) was weighed, and 9 times volume of PBS was added. Clean steel beads were added and placed in a multi-sample grinder (Shanghai Jingxin Industrial Development Co., LTD, JXFSTPRP-24/32) to prepare a tissue homogenate, then the tissue homogenate was centrifuged (DLAB Scientific Co., Ltd. D3024) at 4 $^{\circ}$ C, 3500 rpm for 10 min. The supernatant was aspirated in a new Eppendorf tube for measurement. The levels of inflammatory factors TNF- α , IL-1 β , uPA, CD86, CD206, IL-6, Fizz1 and Ym1 in tissue homogenate supernatants were measured following the kits' instructions. Mouse TNF- α and IL-1 β ELISA kits were purchased from Thermo Biotechnology, USA. Mouse uPA ELISA kit was purchased from Wuhan Cloud-Clone Corp. Mouse anti-CD86, anti-CD206, β -actin antibodies were purchased from Cell Signaling Technology, USA. The primer sequences used for the RT-qPCR reactions are listed in Table 1.

2.5. Immunohistochemical detection of the macrophage phenotypes in lung tissue of ALI mice

Paraffin sections of lung tissue that had been cut into 4 μ m were dewaxed, hydrated with gradient ethanol, and incubated in F4/80, CD86, CD206 primary antibodies (Cell Signaling Technology, USA) at 4 $^{\circ}$ C overnight, washed in TBST, and incubated in HRP-linked anti-rabbit IgG purchased from Abcam. After being developed by diaminobenzidine, the sections were re-stained with hematoxylin to reveal nuclei and observed under an optical microscope (Olympus BX51).

2.6. Real-time fluorescence quantitative PCR to detect the inflammatory factors and uPA in macrophages

RAW264.7 cells were seeded with 1.2×10^6 cells/well in six-well plates. Twenty-four hours later, 1 μ g/mL LPS was given to model group, while cells in Shikonin groups received 0.5 μ M, 1.0 μ M and 1.5 μ M Shikonin (dissolved in 1% DMSO in medium) in the presence of 1 μ g/mL LPS. Same volume of the solvent was given to the normal and model groups. The RNA was extracted for assay after 6 h incubation. RNA was extracted using trizol (Invitrogen, USA) and first-strand cDNA synthesis was performed using PCR kit (TaKaRa, Japan). The mRNA levels of interleukin IL-1 β , IL-6 and TNF- α were assayed by using quantitative PCR with Perfecta SYBR Green PCR kit (TaKaRa, Japan) following previously reported method (Huang et al., 2022). The primer sequences used for the RT-qPCR reactions are listed in Table 1.

2.7. Western blots of inflammatory factors and uPA protein in macrophages

RAW264.7 cells were seeded and treated as described under item 2.6. The total protein was extracted using RIPA buffer (CW2333S, Kangweishiji, China) containing phosphatase inhibitor and protease inhibitors, separated on SDS-PAGE (P0683-500 mL, Beyotime, China), transferred onto a PVDF membrane and incubated in anti-F4/80, anti-CD86, anti-CD206, anti-uPA, β -actin or anti-GAPDH antibodies (Cell Signaling Technology, USA) prior to probe with HRP conjugated secondary antibodies (Cell Signaling Technology, USA). Immunoblots were developed by the chemiluminescence (WBKL S0500, Millipore, USA) and photographed in an imaging station (Tanon-5200, Shanghai Tanon Technology, China).

Table 1
Primer sequences used for PCR reactions.

Primer	Forward (5' – 3')	Reverse (5' – 3')
IL-1 β	GAAATGCCACCTTTTGACAGTG	TGGATGCTCTCATCAGACAG
IL-6	TCTATACCACTTCACAAGTCGGA	GAATTGCCATTGCACAACCTTT
TNF- α	CAGGCGGTGCCTATGTCTC	CGATCACCCCGAAGTTCAGTAG
CD206	CTCTGTTCAGCTATTGGACGC	TGGCACTCCCAACATAATTTGA
Ym1	AGAAGGGAGTTTCAAACCTGGT	GTCTTGCTCATGTGTGAAGTGA
Fizz-1	AGGAGCTGTCTATTAGGGACATC	GGATGCCAACTTTGAATAGG
uPA	GCAGCCATCTACCAGAAGAAACAGG	ACCCAGCAAGGACTGATGAGACTC
uPAR	GAACAATACCCTTGGGTGTTTC	CAGCCTCTTACGGTATAACTCC
MMP9	CGCCACCACAGCCAATATGAC	CTGCTTGCCAGGAAGACGAAG
GAPDH	TTGATGGCAACAATCTCCAC	CGTCCCGTAGACAAAATGGT

2.8. Quantitative proteomics with TMT labeling of RAW264.7 cell samples

RAW264.7 cells received LPS challenge and Shikonin treatments following the protocols listed under item 2.6. The samples were collected, the proteins were extracted and enzymatically digested. Quantitative proteomics was performed by using technologies involving TMT labeling, high-performance liquid chromatography classification, liquid chromatography-mass spectrometry tandem analysis, database search and bioinformatics analysis. Based on the Raw files obtained from the mass spectrometry assay, a sample-specific protein database was firstly constructed, and then the database search was performed using analysis software Maxquant (v1.6.15.0). Then peptide and protein level quality control analysis were performed based on the database search results. The identified proteins were annotated with common functional databases, including GO, KEGG, InterPro, COG, and STRING databases. Next, the proteins were quantitatively analyzed for distribution and reproducibility. Enrichment of KEGG, GO, COG/KOG and protein domain analysis and enrichment-based clustering was performed for the functional classification and further hierarchical clustering of the differences. Fisher exact test was used when there are multiple experimental groups in the project.

2.9. ELISA analysis of the serum uPA levels in normal mice receiving Shikonin

Forty Kunming mice were randomly divided into control group, Shikonin (4 mg/kg, 8 mg/kg or 16 mg/kg) groups (i.p.), and the positive drug tranexamic acid (TXA) group (54.6 mg/kg, i.p.), with 8 mice in each group. Blood samples were collected from each animal 3 h after Shikonin administration, and the serum was centrifuged to detect uPA content following the instruction of uPA detection kit (SEA140Mu, Wuhan Cloud-Clone Corp, China).

2.10. Clotting time test, platelet count test and plasma recalcification time test

To test the influence of Shikonin on clotting time, 40 Kunming mice were randomly divided into control group, Shikonin groups (4, 8 and 16 mg/kg, i.p.) and TXA group (54.6 mg/kg, i.p.), with 8 mice in each group. Three hours after administration, blood samples were collected from the submandibular venous plexus with a drop diameter of about 6 mm, immediately timed with a stopwatch, and gently lifted once every 15 s from the edge of the drop inward with a clean needle to observe the presence of blood clotting. The time interval between the appearance of the blood drop and the thread formation was the clotting time. The time duration of clotting time was confirmed with a parallel blood drop for retesting (Huang, Fan, Yin, & Huang, 2019).

To measure the count of platelets in the peripheral blood of mice, 20 μ L of whole blood was added to 0.38 mL 1 % ammonium oxalate (Solarbio, YS142473) and detected on a veterinary fully automated whole blood analyzer (Shenzhen Mindray Bio-Medical Electronics Co.,

Ltd. BC-2800Vet) (Zhang et al., 2022).

To measure the time of plasma recalcification, 0.1 mL 3.8 % sodium citrate solution was added in 0.9 mL of rabbit blood for anticoagulation (blood: anticoagulant = 9:1). The blood was then centrifuged at 1500 rpm for 10 min, and the upper layer of plasma was aspirated in a clean EP tube. In five 0.5 mL EP tubes containing 0.1 mL of plasma in each tube, 0.1 mL of saline, three doses of Shikonin or positive drug TXA was added in respectively. The mixtures were then incubated at 37 °C for 1 min and then 0.1 mL of 0.025 mol/L preheated CaCl₂ solution (at 37 °C) was added, mixed thoroughly. The tubes were tilted once every 10 s to observe and record the time required for the appearance of fibrin filaments (Zhou et al., 2019). The experiment was repeated 6 times.

3. Results

3.1. Shikonin alleviated acute lung injury, reduced macrophage infiltration and decreased uPA in mice

The microscopic examination of lung tissue stained with H&E revealed the protective effect of Shikonin on mice. Mice in control group had intact lung tissue structure. Neither significant hemorrhage nor inflammatory infiltration was observed. In contrast, mice in ALI model group had obviously damaged lung tissue, thickened alveolar wall, visible hemorrhage, and evident inflammatory infiltration. However, mice receiving low, medium and high doses of Shikonin have statistically alleviated lung injury, reduced hemorrhage, less inflammatory infiltration, and thinner alveolar wall (Fig. 1A), when compared with the model group. The above results demonstrated that Shikonin evidently ameliorated mice ALI.

Macrophage is crucial in mediating inflammatory response in ALI, therefore, the expression of macrophage marker F4/80 and marker of pro-inflammatory polarization CD86 in mouse lung tissues was detected by immunohistochemistry. The results showed that model group has markedly higher F4/80 and CD86 expression but lower CD206 expression compared with control group. While, Shikonin groups have clearly lower F4/80 and CD86 expressions, higher CD206 expression, and their lung microstructures were more similar to that of the control group (Fig. 1B).

The effects of Shikonin on white Blood Cell (WBC), monocyte (Mon) and neutrophil granulocyte (Gran) content in peripheral blood of ALI mice were detected by using a veterinary fully automated whole blood analyzer. The results showed that WBC count, the proportion of monocytes and granulocytes increased significantly in LPS group in comparison to control group. While in Shikonin groups, these indexes decreased significantly in a dose-dependent manner compared to LPS group (Fig. 1C).

ELISA results showed that the levels of IL-1 β , TNF- α and uPA in lung tissues of LPS group were significantly elevated compared with control group. Whereas, mice receiving Shikonin have dose-dependent decreases in IL-1 β , TNF- α and uPA. The most obvious reduction was observed in the high dose Shikonin group (Fig. 2A). Western blot results showed that the levels of CD86 in lung tissues of LPS group significantly

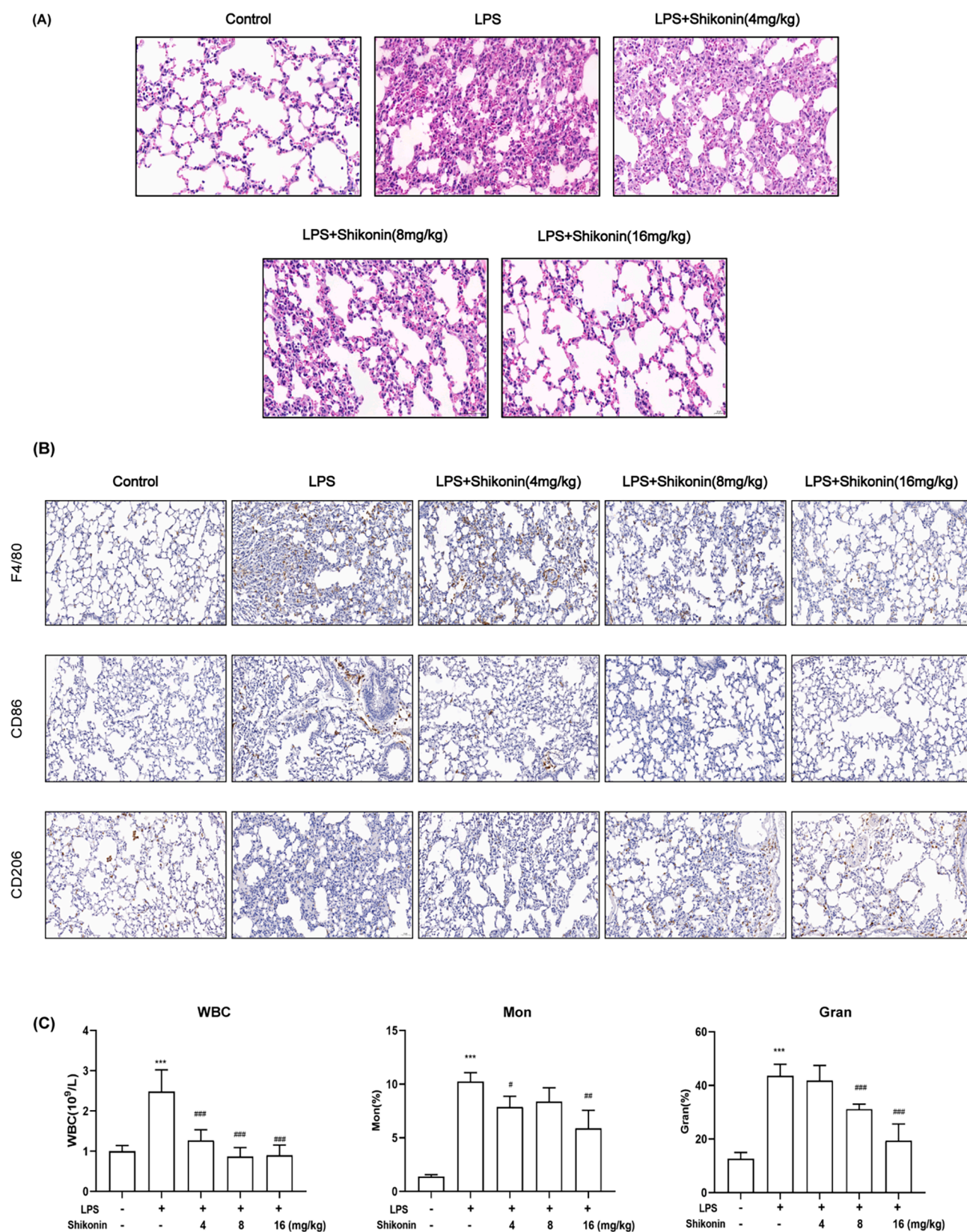
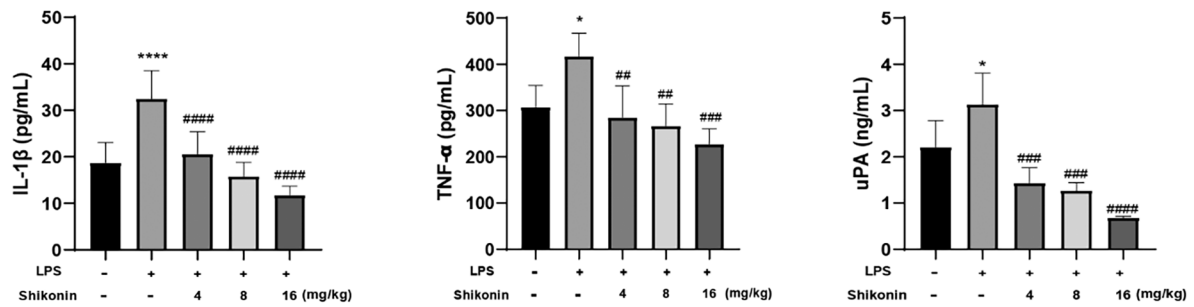
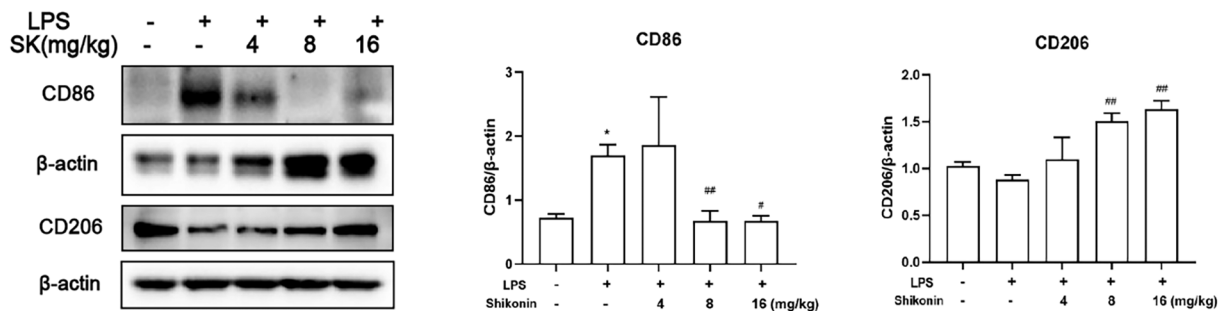


Fig. 1. Shikonin alleviated acute lung injury and reduced inflammation in mice. (A) Histopathological micrographs of lung tissues stained with H&E. (B) Representative immunohistochemistry expression of macrophage marker F4/80, CD86 and CD206 detected in lung tissues. (C) The effects of Shikonin on White Blood Cell (WBC), Monocyte (Mon) and Neutrophil granulocyte (Gran) proportion in peripheral blood of ALI mice.

(A)



(B)



(C)

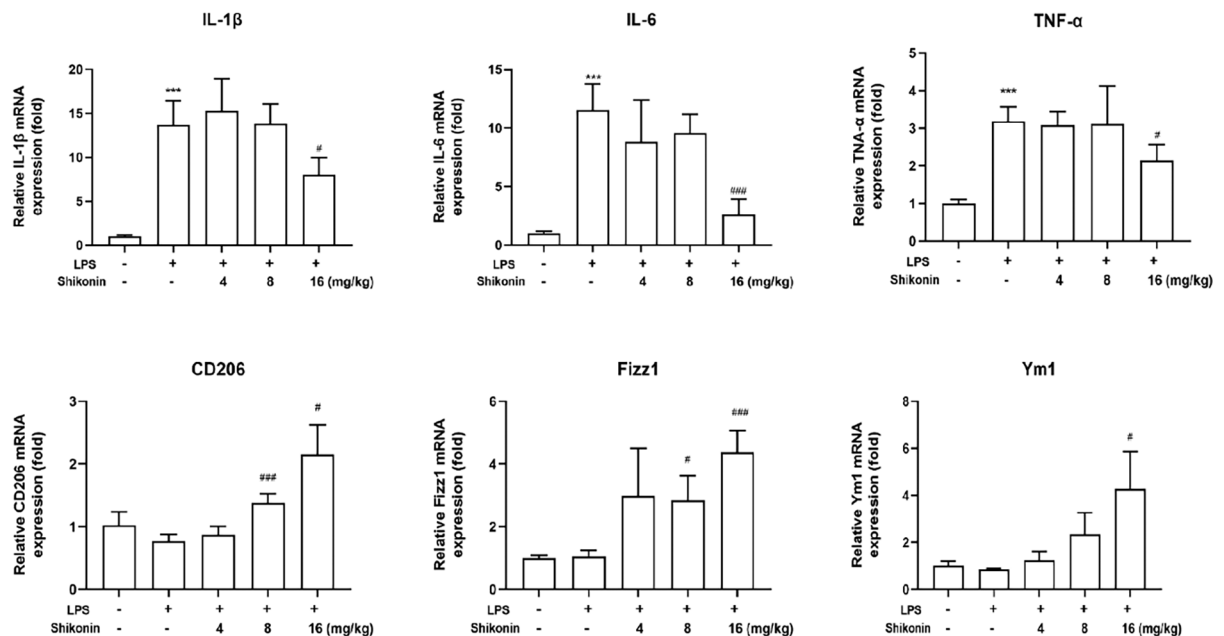


Fig. 2. Shikonin reduced inflammation in ALI mice. (A) The levels of IL-1 β , TNF- α and uPA in lung tissue of mice detected by ELISA. (B) The protein expression of CD86, CD206 and (C) the mRNA expression of IL-1 β , IL-6, TNF- α , CD206, Fizz1 and Ym1 in the lung tissues of LPS-induced ALI mice.

raised compared with control group. While in Shikonin groups, the levels of CD86 significant decreased and the levels of CD206 apparently elevated (Fig. 2B). Consistently, the mRNA expression of IL-1 β , IL-6 and TNF- α increased notably in LPS group compared with control group. However, the expression of these genes decreased in Shikonin group. In addition, compared with the LPS group, the expression levels of CD206, Fizz1 and Ym1 genes in the lung tissues of ALI mice were significantly increased in Shikonin groups (Fig. 2C).

3.2. Shikonin reduced LPS-induced inflammatory factor and uPA expression in macrophages

To further investigate whether Shikonin ameliorated LPS-induced lung injury via restraining macrophage-mediated inflammation, we conducted in vitro experiments on RAW 264.7 derived macrophages. The mRNA expression of IL-1 β and TNF- α in LPS-challenged macrophages was dose-dependently reduced in all Shikonin groups, compared

with that of LPS group (Fig. 3A-B). Besides, both the mRNA and protein expression of uPA are significantly reduced in Shikonin groups in comparison to LPS-challenged macrophages (Fig. 3 C-D). As MMP-9 is one of the most important down-stream signals of uPA, its' expression was detected by q-PCR. The result showed that Shikonin significantly inhibited the LPS-induced mRNA expression of MMP-9 in macrophages (Fig. 3 F). However, Shikonin showed no influence on uPAR expression (Fig. 3 E). The above results indicated that Shikonin significantly reduced LPS-induced inflammatory factors and uPA in macrophages, but may not affect uPAR.

3.3. Effect of Shikonin on protein expression in LPS-primed macrophages

To further investigate the effect of Shikonin on protein expression in LPS-primed macrophages, hierarchical clustering calculations of differentially expressed protein quantification and protein dimension were performed as shown in Fig. 4A. The differentially expressed proteins among the three groups were screened out, by comparing the expression levels of total proteins. The increment more than 1.3-fold was regarded as significant up-regulation and the decline more than 0.8-fold was regarded as significant down-regulation. In Fig. 4B, red dots denote up-regulated differential protein, green dots represent down-regulated differential protein, and grey dots are proteins without difference. Difference analysis revealed that there were 1905 differential proteins between the control and model groups, 58 differential proteins between the Shikonin and LPS groups, and 36 communal differential proteins were identified in both comparisons.

Among the 36 differential proteins, protein expression of uPA, CD40, IKB- α , IL1- β , and IL-6 in macrophages was observed to be increased in the LPS group and significantly decreased in the Shikonin group (Fig. 4B). Orange dots in the graph indicate proteins with significant differential expression that are upregulated. Green dots represent proteins with significant differential expression that are downregulated, and gray dots denote proteins without differences.

The gene ontology (GO) approach integrates intra- and inter-species representations of genes and genes products, which serves to annotate genes and their products in terms of their molecular functions, cellular locations, and biological processes. GO analysis revealed that these proteins are involved in: 1) biological processes (BP) such as cellular processes, biomodulation, response to stimuli, and immune system processes; 2) cellular component (CC) such as cells, intracellular regions, and protein-containing complexes; 3) molecular functions (MF) such as binding, molecular function management mechanism, catalysis, transport, and molecular transduction. (Fig. 4C).

Two important functional pathways were identified in this analysis. One pathway is associated with NF-kappa B signaling and includes 4 down-regulated proteins, namely, uPA, CD40, IKB- α and IL1- β . The other pathway is associated with complement and coagulation cascades, in which, likewise, uPA was also found in the down-regulated proteins (Supplementary Fig. 1). The above results demonstrated that Shikonin alleviated the inflammatory response in LPS-primed macrophages and regulated the protein expression of various proteins involving uPA.

3.4. Shikonin accelerated blood coagulation and decreased uPA in mouse serum

The above results demonstrated that uPA may involve in the anti-inflammatory effect of Shikonin. It is well known that uPA inhibited blood coagulation by catalyzing fibrinolysis. Thus, Shikonin may mediate a hemostasis effect via inhibiting uPA. The in vivo results showed that Shikonin significantly shortened coagulation time in mice, increased platelet count, and reduced plasma recalcification time in rabbits (Fig. 5A-C). Furthermore, mice given Shikonin had dose-dependent lower serum uPA level compared with control group. The uPA in high dose Shikonin group was even lower than that of TXA group (Fig. 5D). The above results showed that Shikonin had significant

hemostatic efficacy and clearly reduced uPA in animal serum.

4. Discussion

ALI is an acute and life-threatening respiratory disease characterized by severe inflammation and organ failure. LPS challenge in mice can induce monocyte recruitment and conversion into lung macrophages, which are typical characteristics in human ALI. Macrophages in pulmonary tissues act as critical and versatile effectors to initiate host defense and to enable tissue degradation and remodeling. In this experiment, we found that LPS induced mice ALI and macrophage pro-inflammatory recruitment in lung tissue. The protein expression of CD86, a marker of macrophage pro-inflammatory polarization, as well as proinflammatory cytokines IL1- β , TNF- α and IL-6 were elevated in ALI model group. Concurrent with the increased proinflammatory cytokines, the mRNA and protein expression of uPA in lung tissue were also significantly elevated. This is consistent with earlier reports in ALI mice (Nassar et al., 2011).

LPS are known to prime macrophage pro-inflammatory polarization, which rely on membrane-bound proteases including MMPs and uPA/uPAR for extracellular matrix degradation, and uPA promotes LPS-induced leukocyte infiltration (Kim et al., 2012; Sugioka et al., 2014). As a serine protein hydrolase, uPA participates in diverse physiologic and pathophysiologic processes. It's proteolytic activity not only enables fibrinolysis, but also primarily associates with degradation of ECM. Therefore, uPA is believed to be a central regulator of macrophage 3D invasion, matrix remodeling, and adhesion, and plays a key role in multiple levels of the inflammatory response (Fleetwood et al., 2014). Besides, uPA was believed to increase pulmonary arterial contractility, impair permeability after ALI, while lung macrophages were the primary reservoir of uPA after cardiopulmonary infection (Bondu et al., 2018; Nassar et al., 2011). In patients with acute and chronic inflammatory conditions, involving pneumonia and severe infection, elevated levels of tissue and circulating uPA are present for prolonged periods, which are often accompanied by the development of ALI (Nassar et al., 2011).

In this study, as is speculated, Shikonin alleviated the symptoms of ALI, decreased inflammatory factors, and inhibited the infiltration of macrophages in the lung tissue of ALI mice. These effects are consistent with previous reports (Zhang et al., 2018). RAW264.7, as a classical monocyte macrophage cell line, has been extensively used to study macrophage functions, mechanisms, and signaling pathways. Study on cell line is more convenient and economic compared with separation primary macrophages from animals. Therefore, LPS-induced RAW264.7 are extensively used in previous literatures to clarify the anti-inflammatory mechanism of potential pharmaceutical candidates, in combination with in vivo data on LPS-induced mice ALI (Cai, Wang, Sheng, Zhang, & Lv, 2022; Li et al., 2023; Wang et al., 2022). As Shikonin is the major active ingredient in *Arnebiae Radix*, a representative herb with circulation invigorating, anti-inflammation and hemostatic effects (Y. Zhang et al., 2018), it can be speculated that Shikonin's inhibition on uPA possibly is a communal foundation enables the various of activities. This speculation is firstly supported by the down-regulated expression of CD86, uPA, TNF- α , IL1- β and IL-6, while increased CD206 in lung tissue of ALI mice. This effect of Shikonin was further confirmed by proteomic results revealing that Shikonin reduced protein expression involving uPA, CD40, IKB- α , IL1- β , and IL-6 in LPS-primed macrophages. Furthermore, in vivo and in vitro coagulant tests showed that Shikonin has a hemostatic effect. Shikonin increased platelet count, shortened coagulation time, and dose-dependently decreased serum uPA level in mice. It also reduced plasma recalcification time in rabbit plasma.

These results collectively suggest that the inhibitory effect on uPA is possibly an important mechanism contribute to the anti-inflammation and hemostatic effect of Shikonin. Besides, current data suggest that Shikonin did not influence the mRNA expression of uPAR. However, beside of uPA, the fibrinolytic system also includes other components,

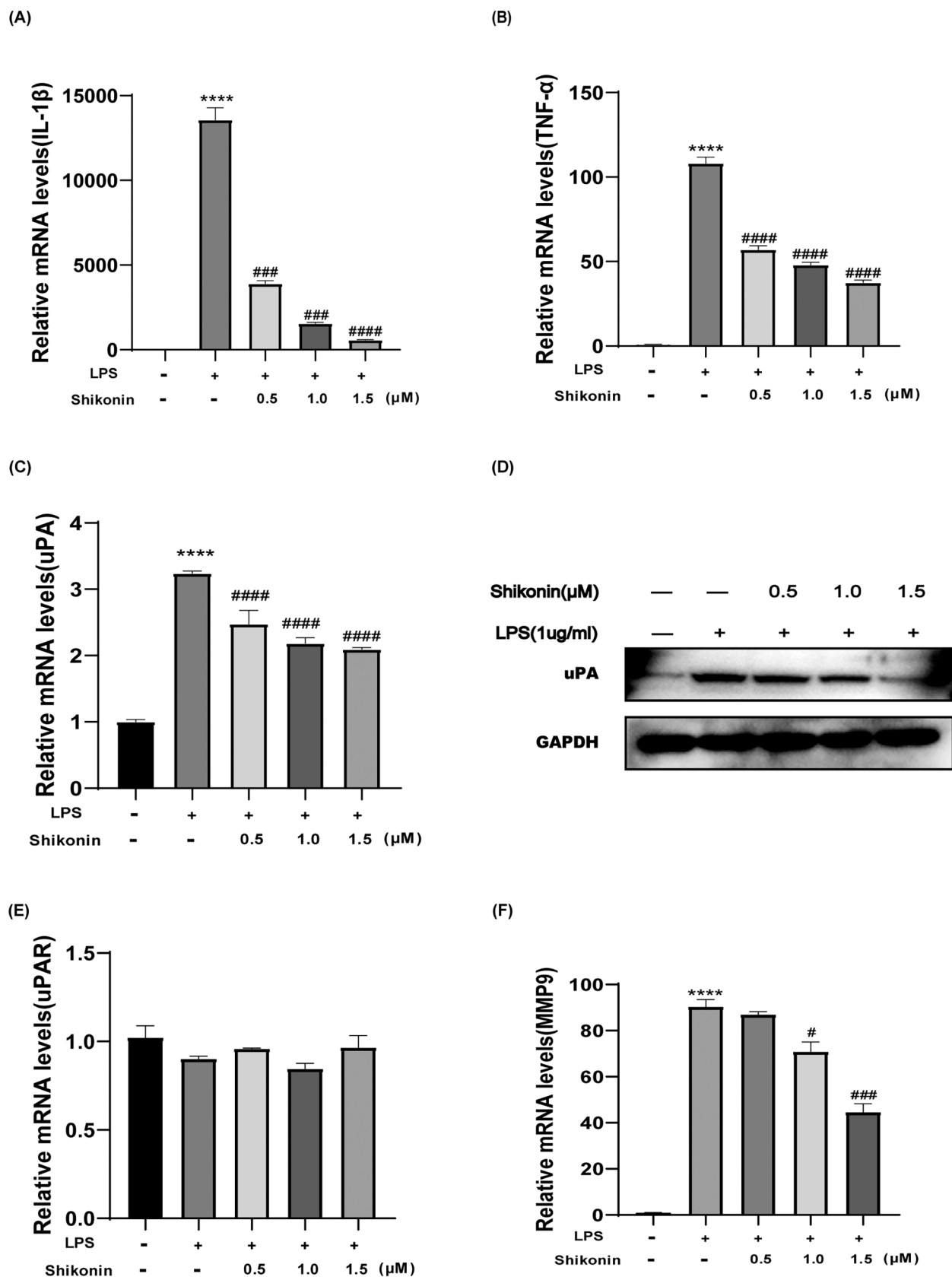


Fig. 3. Shikonin reduced LPS-induced inflammatory factor and uPA expression in macrophages. The mRNA expression of IL-1 β (A), TNF- α (B) and uPA (C) in macrophages. The protein expression of uPA (D), the mRNA expression of uPAR (E) and MMP-9 (F) in LPS-primed macrophages.

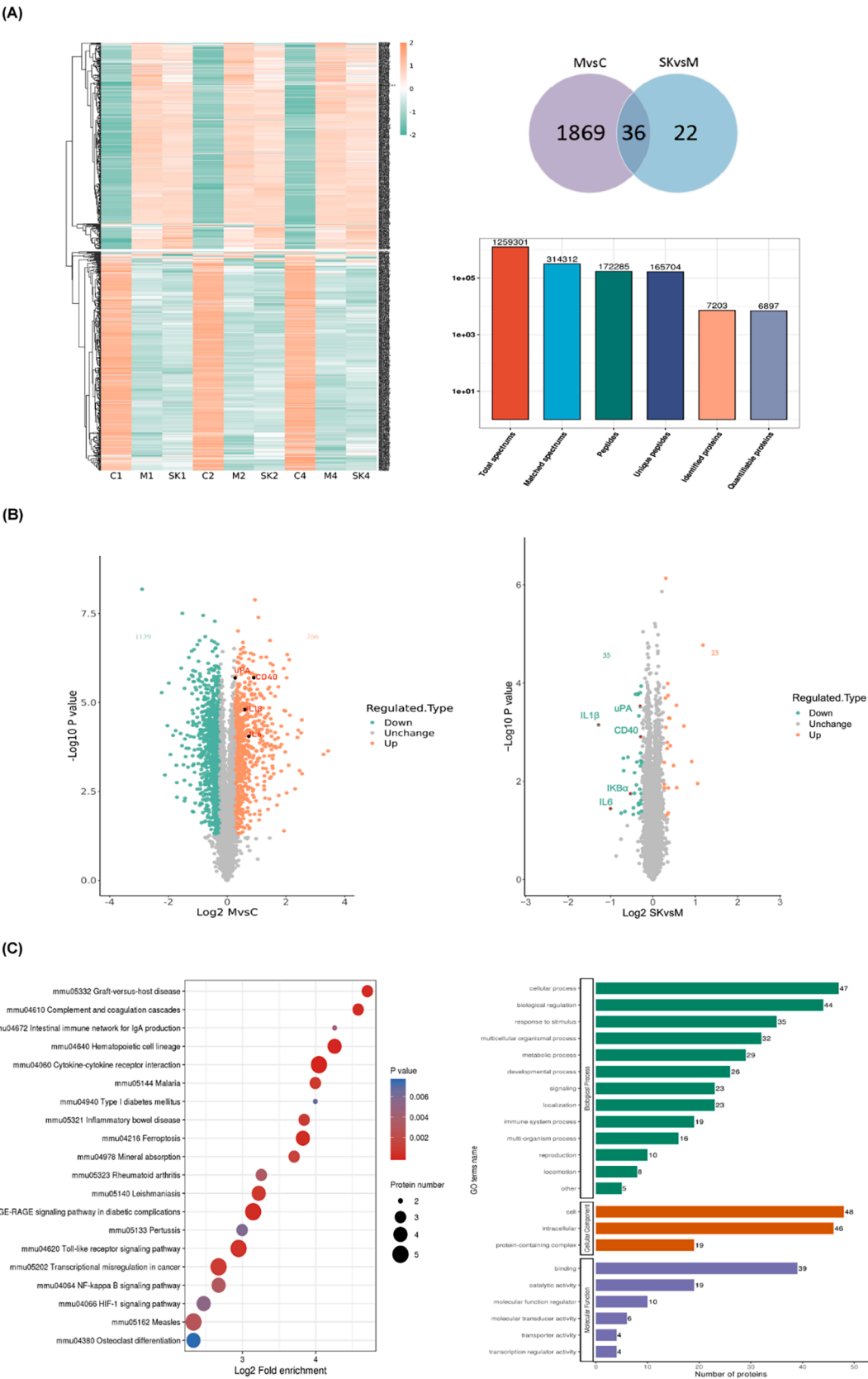


Fig. 4. Shikonin reduced protein expression involving IKB- α , IL1- β and uPA in LPS-primed RAW-264.7 cells. (A) Hierarchical clustering plot of proteomic results, differential protein Wayne plot, and informative statistical plot of protein identification. (B) Volcano plot of differential proteins, (C) Bubble plot of differential protein KEGG enrichment analysis and bar graph of GO enrichment analysis.

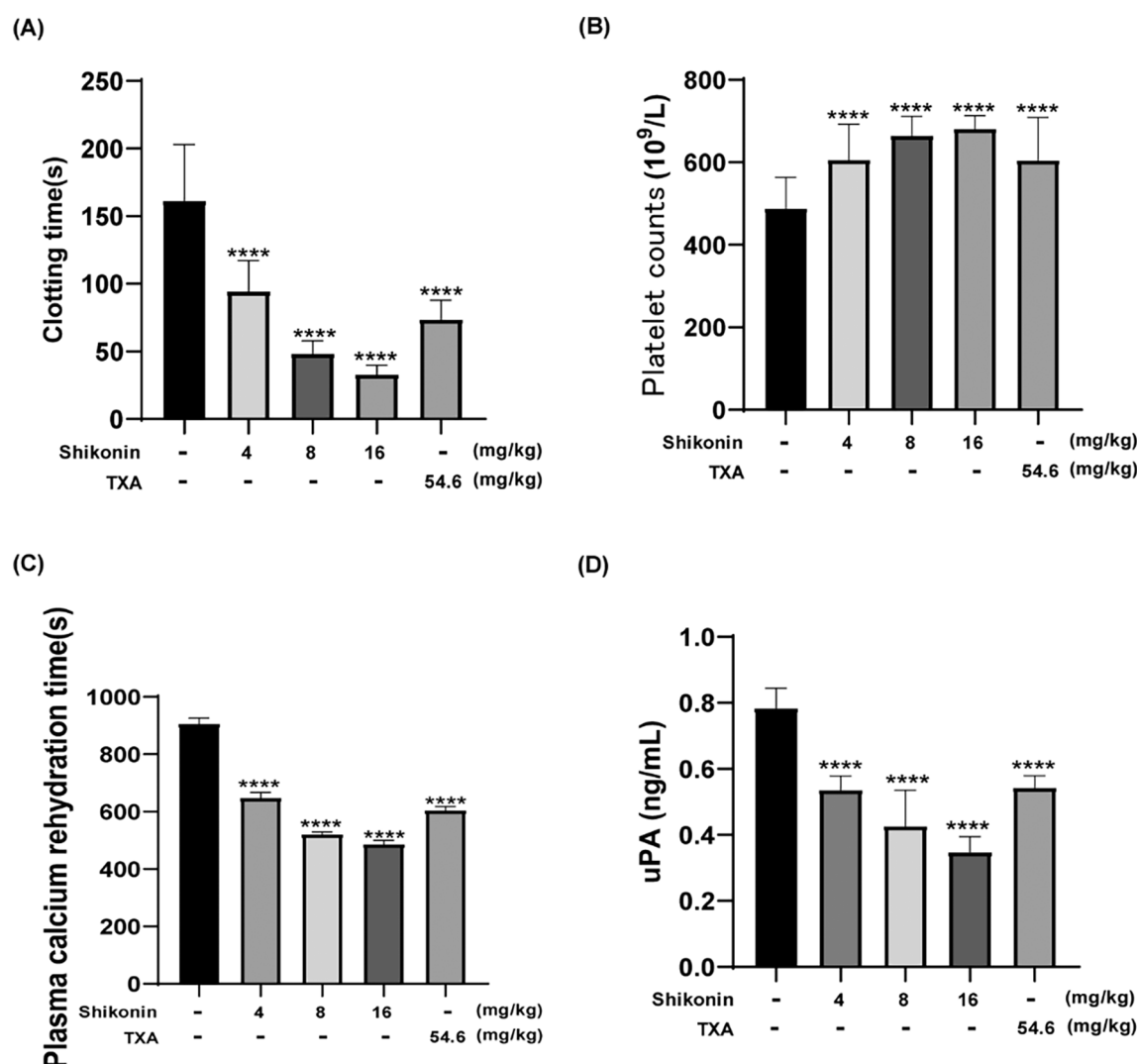


Fig. 5. Shikonin reduced clotting time (A), increased platelet count (B), shorted plasma recalcification time (C), and decreased the level of uPA (D) in mice serum.

for example, tissue plasminogen activator (tPA), anti-fibrinolytic components α 2-Antiplasmin, PAI-1 and thrombin-activated fibrinolytic inhibitor (TAFI) (Franchini, Zaffanello, & Mannucci, 2021). Whether Shikonin influences these factors remains to be further validated.

Another characterization of ALI is the accumulation of activated neutrophils, which clearance from lung plays a crucial role in resolution of this pathophysiologic process. Previous data showed that the phagocytosis of apoptotic neutrophils by macrophages was significantly inhibited by uPA, suggesting uPA to be a therapeutic target of ALI (Y. Yang et al., 2010). In this study, Shikonin significantly decreased the accumulation of both neutrophils and macrophages in ALI mice, yet whether Shikonin promotes efferocytosis still merits further exploration.

5. Conclusion

This study consolidated the anti-inflammatory effects of Shikonin on LPS-induced mice ALI and LPS-primed macrophages. The results suggested that the anti-ALI and hemostatic effects of Shikonin closely related to the inhibition on uPA. These data provide preliminary evidence supporting the clinical application of Shikonin and *Arnebiae Radix* on ALI and hemorrhagic disorders.

6. Ethics statement

All animal experiments were performed on the basis of the National

Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH Publications No. 8023, revised 1978) with approval from the Animal Experiment Center, Guangzhou University of Chinese Medicine (license No. SCXK, Guangdong 2019-0047).

Funding

This research was funded by National Natural Science Foundation of China (No.82074382), National Natural Science Foundation of China (No. 8207142520), and Natural Science Foundation of Guangdong Province (No. 2021A1515011232).

CRediT authorship contribution statement

Chengcheng Li: Methodology, Data curation, Investigation, Writing – original draft, Validation. **Shuru Lu:** Methodology, Validation. **Chunqi Xu:** Writing – original draft, Investigation, Validation. **Baoyuan Huang:** Investigation, Validation. **Qiumei Wang:** Methodology. **Enxin Zhang:** Project administration, Conceptualization, Supervision. **Xiao-jun Zhang:** Project administration, Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jff.2023.105527>.

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